

Technical Bulletin Unique Identifier	DOC2919091
Title	Service Note for introduction of new Low Height Fixed Table Home Sensor Mechanism kit 8311111 and its replacement
Subject	Introduction of new Low Height Fix Table Home Sensor Mechanism kit and its replacement
Applicability (e.g., affected product, process)	1.5T SIGNA Creator/Explorer 1.5T SIGNA MR355/MR360 1.49T SIGNA MR380 1.5T SIGNA Prime 1.5T SIGNA Victor/ Star Air /Aviator Air
Date	19-Mar-2024
Relevant Product Business (select one)	☒ Imaging ☐ Patient Care Solutions ☐ Ultrasound ☐ Other: _____
Originating Modality	MR
Details	Introduction of new Low Height Fix Table Home Sensor Mechanism kit and its replacement
Tracking Number	ECR2370402/ MR2354913
Affected Service Publications	Signa Creator/Explorer 1.5T System Service Manual (Class A: 556356-2EN Rev.10, Class C: 5756355-2EN Rev.11, Class M: 5756347-2EN Rev.11) Signa MR 355/MR 360 1.5T System Service Method (Class A: 5856356-3EN Rev.6, Class C: 5856355-3EN, Rev.7 Class M: 5856347-3EN Rev.6) Signa MR 380 1.49T System Service Manual (Class A: 5856357-8EN Rev.3, Class C: 5856358-2EN Rev.2, Class M: 5856359-8EN Rev.2) 1.5T SIGNA Prime Service Manual (Class A: 5807565-2EN Rev.4, Class C: 5807569-2EN Rev.4, Class M: 5807568-2EN Rev.3) 1.5T SIGNA Victor/SIGNA Star Air/SIGNA Aviator Air Service Manual (Class A: 5957356-8EN Rev.1, Class C: 5957355-8EN Rev.1, Class M: 5957347-8EN Rev.1)
Resolution	See attached
Technical Bulletin Owner	Qin Yu (SSO:503343456)

(This SN is classified as Class A)

Effectivity:

1.5T SIGNA Creator/Explorer
1.5T SIGNA MR355/MR360
1.49T SIGNA MR380
1.5T SIGNA Prime
1.5T SIGNA Victor/ Star Air /Aviator Air

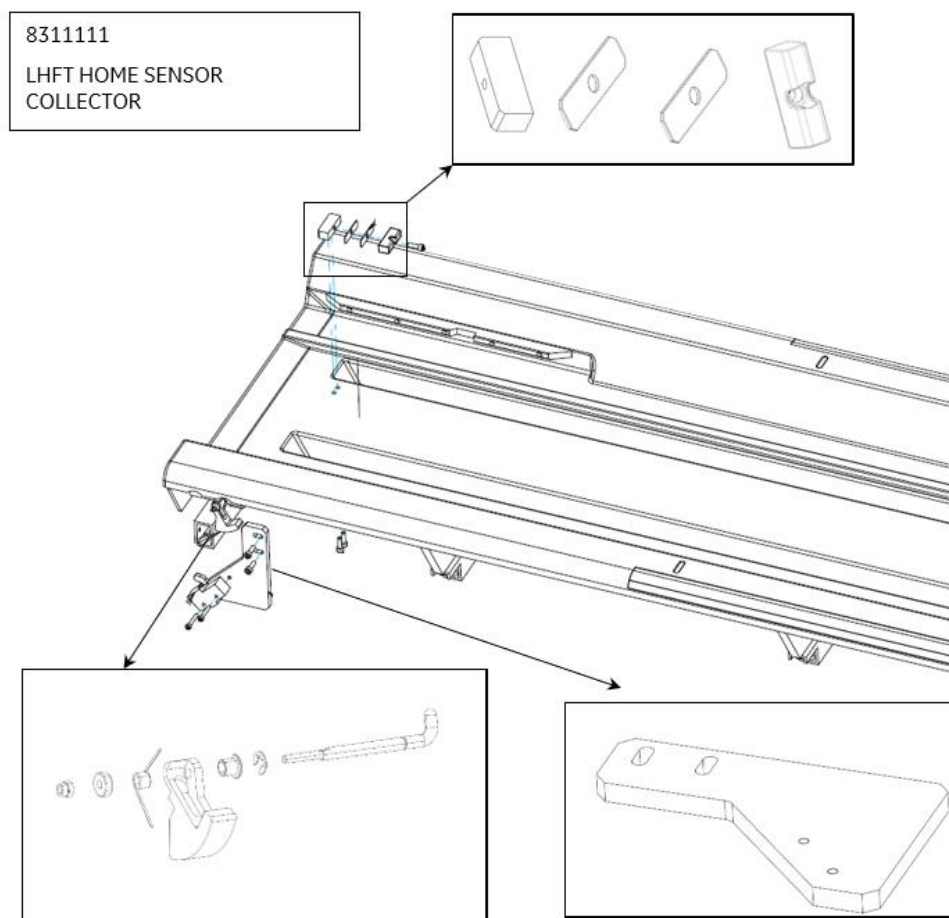
Overview:

LHFT home sensor mechanism improvement on Tabletop parts 5440648-3 & 5440648-2

Due to home sensor pin bend which lead to home limit sensor (46-136334P49) is not activated and cannot move table downwards. LHFT home sensor mechanism improvement has been made and set new low-level FRU Home Sensor Mechanism Collector kit (8311111) for Tabletop parts 5440648-3 & 5440648-2.

NOTE: Up/Home Limit Sensor (46-136334P49) is not included in LHFT HOME SENSOR Mechanism COLLECTOR kit (8311111).

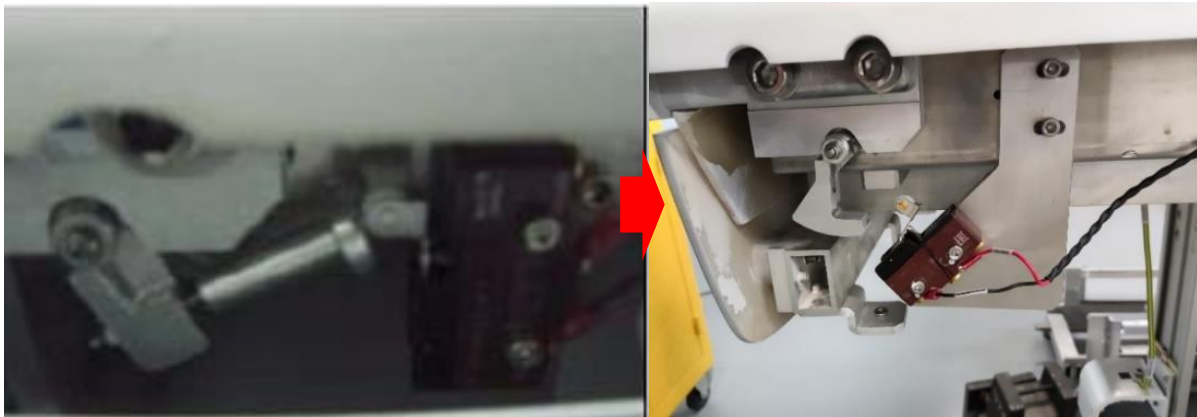
Figure 1. 8311111 LHFT HOME SENSOR Mechanism COLLECTOR



This Service Note introduces new Home Sensor Mechanism Collector kit (8311111) and its replacement procedure for old home sensor mechanism part (5445684) and new home sensor mechanism part (8311111).

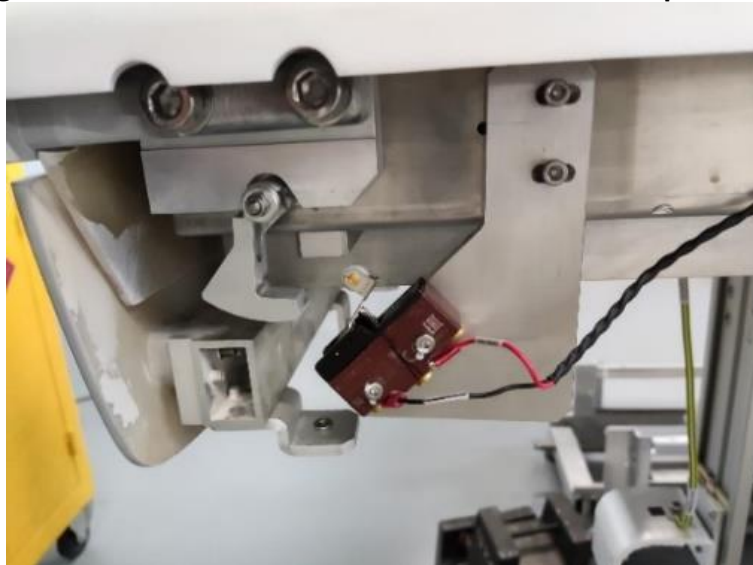
Old home sensor mechanism part (5445684) upgrade replaces by new home sensor mechanism kit (8311111):

Figure 2. Old home sensor mechanism part (5445684) upgrade replaces by new home sensor mechanism kit (8311111)



New home sensor mechanism kit (8311111) replacement:

Figure 3. New home sensor mechanism kit (8311111) replacement



FRU list update for service manual:

(The Yellow parts is new updated)

Figure 4. Tabletop ASSY and related FRU parts

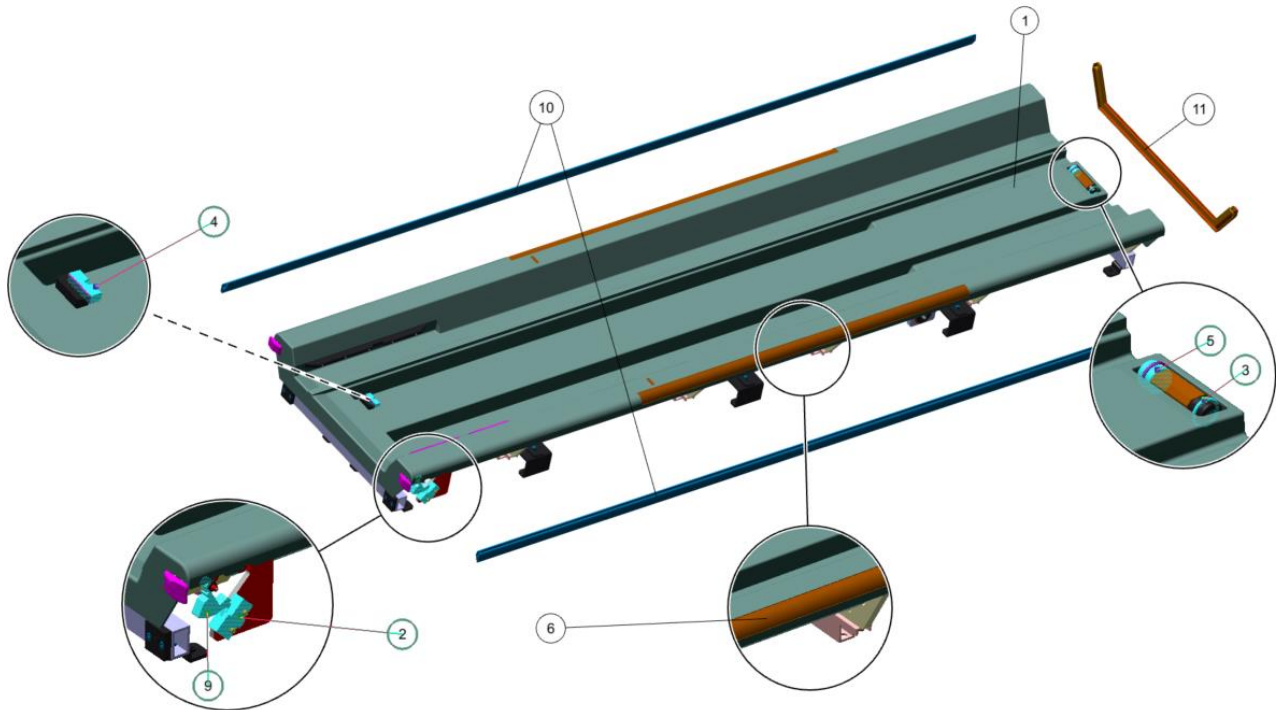


Table 1. Tabletop ASSY and related FRU parts

Item	Part Number	Description	Quantity
9	5445684	Home Sensor	1
	8311111	LHFT HOME Mechanism SENSOR KIT	1

Add Section: “Low Height Fix Table Home Sensor Mechanism kit” to service manual:

Prerequisites:

Table 2. Personnel requirements

Required persons	Preliminary requirements	Procedure	Finalization
1	-	1 hours	-

Table 3. Tools and test equipment

Item	Quantity	Part number	Manufacturer
Allen Key Set - Metric	1	5115275	-
Torque Wrench, 8-50 N m Adjustable, Nonmagnetic	1	5534134 or 5534134-2	-
Spanners (1/2,3/4,3/8,7/16,5/16,32)	One for each size	-	-
Circlip inserter (A-150,A170)	One for each size	-	-
Hex. Socket (9/16,3/4,3/8)	One for each size	-	-
Scale (15cms & 30 cms)	One for each size	-	-
Ball driver (5/32,5/64,3/32,1/8,7/64,9/16)	One for each size	-	-
Flat nose plier	-	-	-
Cutter	-	-	-
Copper hammer	-	-	-
Screwdriver (8mm, 4mm)	One for each size	-	-
Knife	-	-	-
Soft mallet	-	-	-

Table 3. Replacement parts

Item	Quantity	Part number	Manufacturer
LHFT HOME SENSOR Mechanism COLLECTOR	1	8311111	-

Table 4. LHFT HOME SENSOR Mechanism COLLECTOR 8311111 Kit content

Part number	Description	Qty	Actual pic
8311130	LHFT Home sensor mount	1	
5366148-3	BUMPER	1	
8311142-1	Position Measurement Sticker of LHFT	4	
8311133	CRADLE STOPPER	1	
5371147-3	Shim	4	

8312222	LHFT HOME SENSOR KIT	1	See below
5331025	Flange Bearing WFM-0608-08	1	
8311131	LIMIT SWITCH DOG & HOME SENSOR TRIGGER	1	
5366566	Spacer	1	
8311137	LHFT Home Sensor Torsion Spring	1	
8311135	LHFT Home Sensor Circlips 5mm	1	
8311134	LHFT HOME SENSOR PIN	1	
3002-M4C-33	PREVAILING TORQUE TYPE HEXAGON NUT WITH NON-METALLIC INSERT, ISO 7040, M4-0.7 A4-70 STAINLESS STEEL, RoHS FASTENER	1	

Table 5. Safety

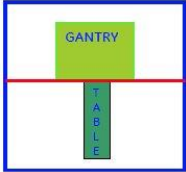
Before working in any GE Healthcare MR suite or performing any GE Healthcare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE Healthcare MR Service Safety Manual (5452735). • Have successfully completed all relevant GE Healthcare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.	
WARNING	
	STRONG MAGNETIC FIELD Ferrous materials can become dangerous projectiles in the presence of the magnetic field that the magnet produces. Do not bring any ferrous tools or equipment into the magnet room.
NOTICE	
POSSIBLE COMPONENT DAMAGE The force of the magnetic field of the MR system can cause the patient table actuator not to function. Do not take the table beyond the marked line as shown below. 	

Table 6. Required conditions

Table is at topmost position.

Procedure:

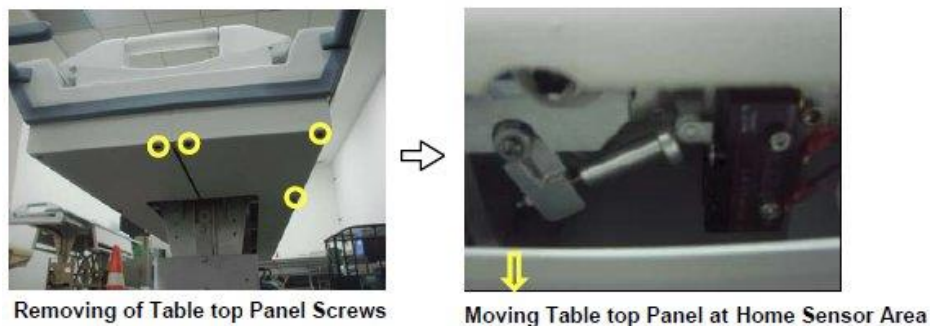
For below steps please refer to from Service Manual:

1. Release cradle from LPCA
2. Applying LOTO-ISC
3. Removing FRP bottom covers
4. Removing dock cover assembly
5. Removing scissor covers

Remove home sensor and related parts:

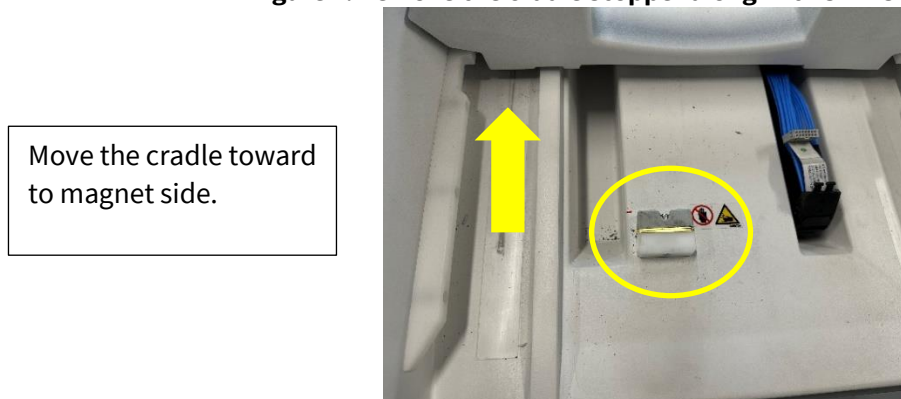
6. Remove the right panel screws and remove the right panel to access to the home sensor.

Figure 1. Remove the right panel screws and remove the right Panel



7. Push the cradle towards the magnet side to access cradle stopper. Remove cradle stopper along with shims and bumper by remove screws.

Figure 2. Remove the cradle stopper along with shims and bumper



8. Open all fixer of cable that connect with home sensor.

Figure 3. Open all fixers



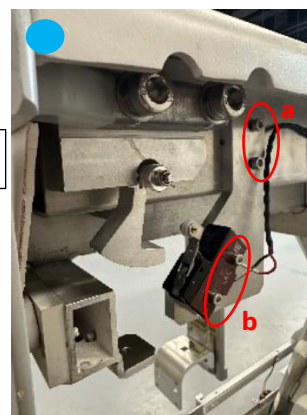
Open all fixers on
the table right side

9. Remove the home switch mounting screws, then removed home switch from original sheet metal. (For old home sensor trigger to new home sensor, please refer figure with green mark, for new home sensor trigger replacement, please refer figure with blue mark.)

Figure 4. Remove mounting screws



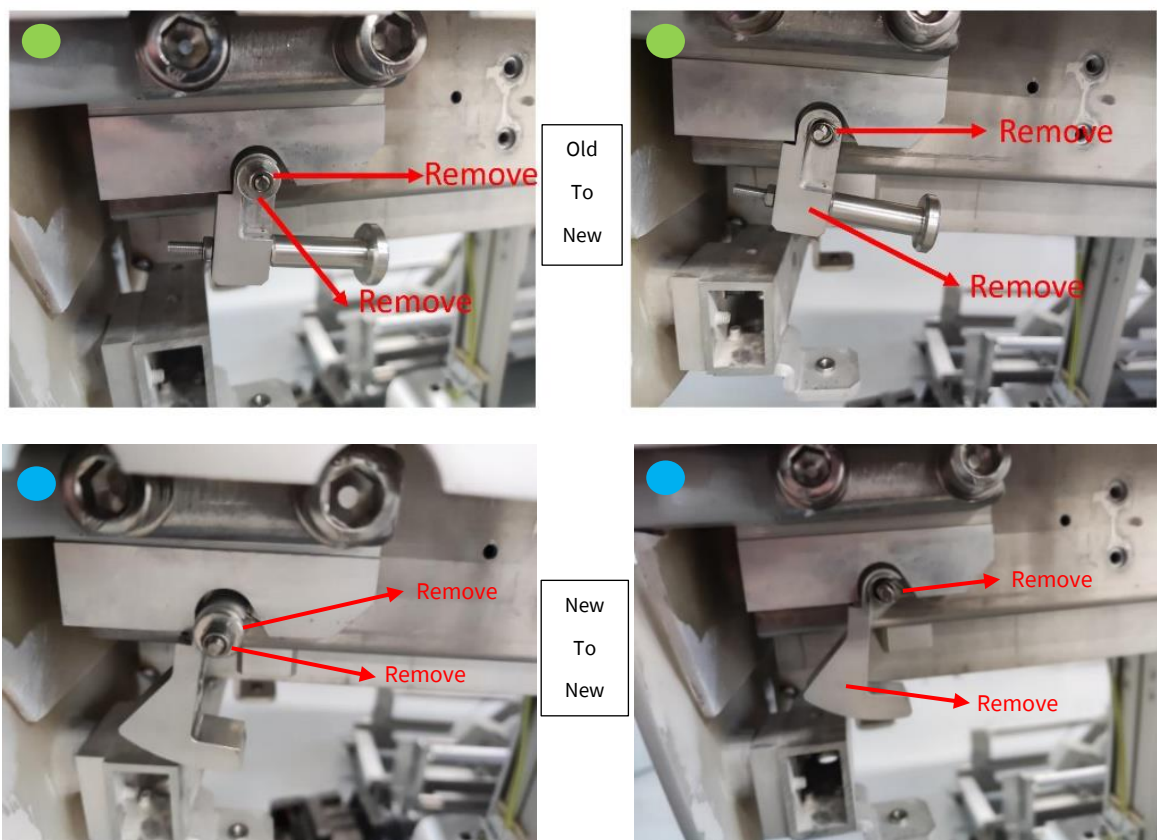
OR



Removal of sensor Limit Switch (old or new)

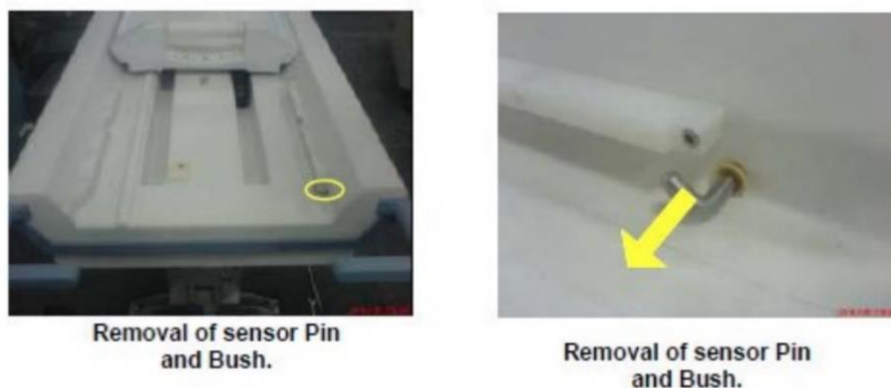
10. Remove the limit switch dog nut, and then remove the washer and limit switch dog along with the spring (For old home sensor trigger to new home sensor trigger, please refer figure with green mark, for new home sensor replacement, please refer figure with blue mark.)

Figure 5. Remove the limit switch dog, spacer, nut and spring



11. Remove the sensor pin, circlip, and the flange bush from the other side of the Tabletop.

Figure 6. Removal of sensor Pin and Bush

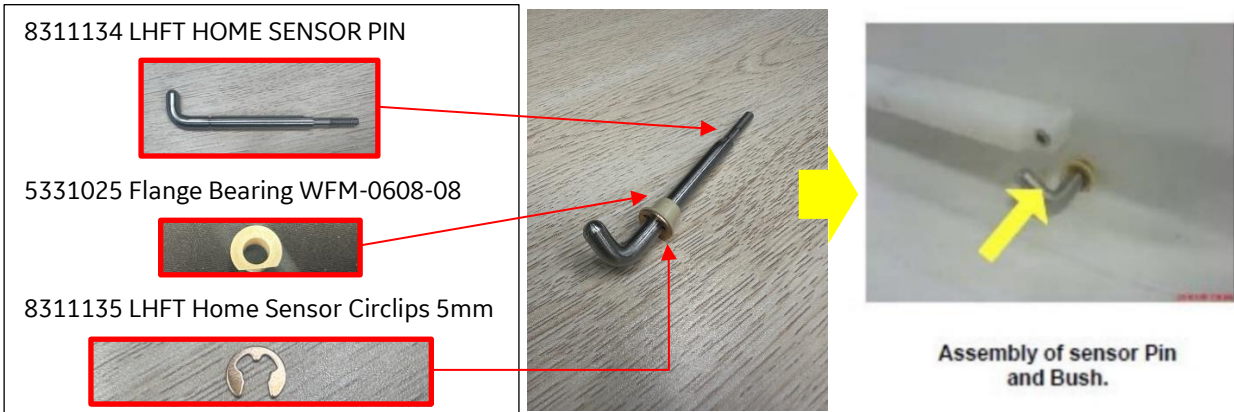


Install New LHFT HOME SENSOR Mechanism Collector (8311111):

12. Now take the parts from the new LHFT HOME SENSOR collector (8311111), assemble PIN (**8311134 LHFT HOME SENSOR PIN**) into the bush (**5331025 Flange Bearing WFM-0608-08**), and insert the circlip (**8311135 LHFT Home Sensor Circlips 5mm**) to sensor rod. Then put them into the hole.

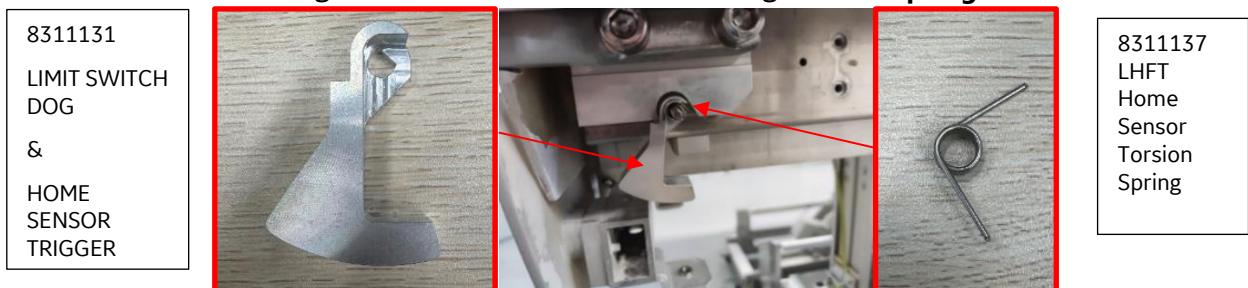
NOTE: Please adjust to ensure that the flat edge of the bush and the opening direction of the clipper are consistent and fit the inside edge of the table board.

Figure 7. Assemble the Sensor rod into the Bush and install clipper



13. Insert the new limit switch dog (**8311131 LIMIT SWITCH DOG & HOME SENSOR TRIGGER**) to the sensor pin, then insert the spring (**8311137 LHFT Home Sensor Torsion Spring**) from the collector into the limit switch dog.

Figure 8. Insert the new limit switch dog and the spring



14. Fix the washer and lock nut to the sensor pin locking the switch dog. Ensure spring action of the switch dog by actuating it.

Figure 9. Fix Limit switch dock lock with spacer and nut0



15. Fix the home switch with mounting screws (reuse from **Step.10**) to the sensor mount (**8311130 LHFT Home sensor mount**).

Figure 10. Fix LHFT Home sensor mount with screw (reuse from Step.10)



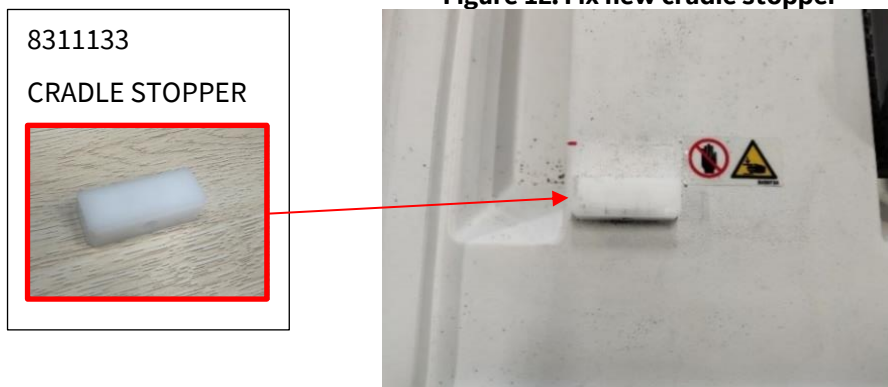
16. Install new home sensor mount from collector at standard middle position (reuse from **Step.7** screw, do not too tightly, may need to adjust the position later).

Figure 11. Fix new home sensor mount



17. Install new cradle stopper (**8311133**), the side without chamfer is directed toward the magnet (reuse from **Step 8**).

Figure 12. Fix new cradle stopper



18. Install 2 new shims and new bumper with the screw from **Step.8** to cradle stopper.

NOTE: New shim is thicker than before, new bumper is harder than before.

NOTE: Tighten the screw on bumper until it's fixed, do not over tighten.

Figure 13. Install new shims



Calibrate the cradle position:

19. Verify whether side latch will cause scratch, if not, go to next step, if yes, adjust side bar vertically to ensure gap between cradle & side bar

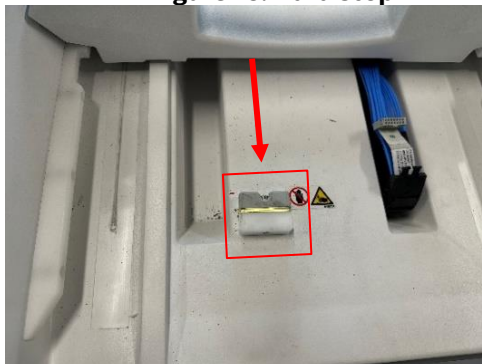
- 1) Loose screws to adjust side bars vertically, tighten after adjustment.
- 2) Push cradle back to home position, check gap between cradle and side bar, ensure no scratch when cradle goes.

Figure 14. Verify whether side latch will cause scratch



20. Pull the cradle back to cradle stop (**Figure 15**), then push the cradle forward to magnet side, confirm whether it can be locked by the side latch.

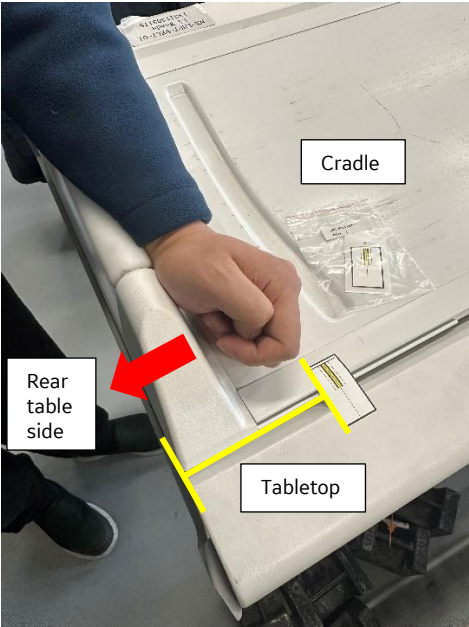
Figure 15. Hard stop



***NOTE:** This step is to measure the shaking distance of cradle hard stop to side latch
If the cradle can be locked by the side latch, keep push the cradle towards the cradle hard stop, place a position measurement sticker (**8311142-1 Position Measurement Sticker of LHFT**) between cradle and tabletop (rear table side).

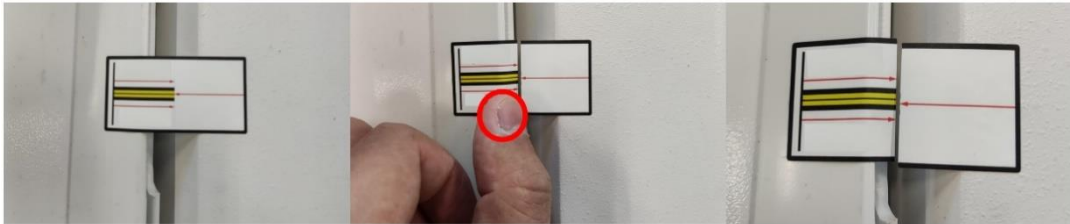
NOTE: Place the sticker parallel to the end of the table.

Figure 16. keep push the cradle, place the sticker



Use finger push the middle of the sticker to cut the sticker between cradle and tabletop.

Figure 17. Using finger to cut the sticker

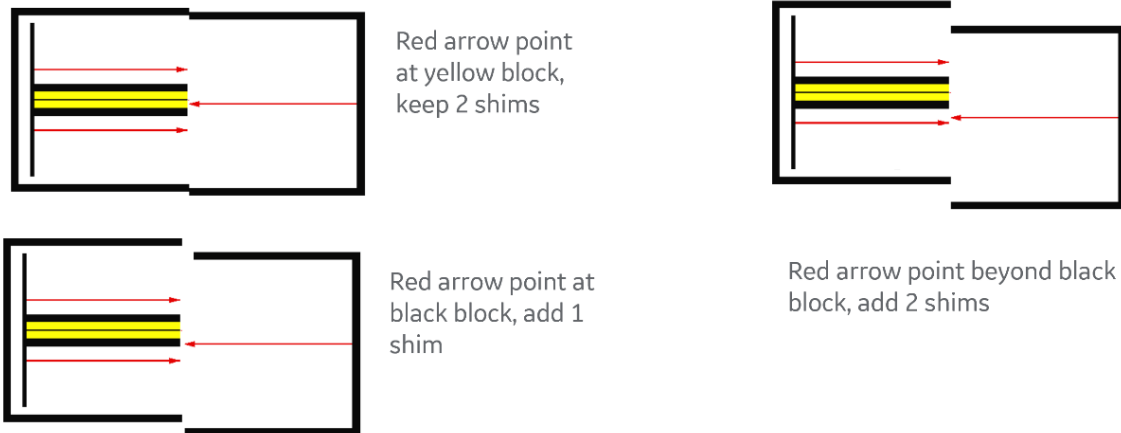


Move the cradle towards to magnet side until it locked by side latch, measure the distance between position measurement stickers, follow the **Figure 18** to adjust the number of shims (back to step), double check the cradle can be locked by side latch.

NOTE: The number of shims in the figure below defaults is 2 shims.

Figure 18. Use the sticker to calibrate the cradle position

Center arrow pointed position	Yellow	Black	Beyond Black
Amount of shims	2	3	4

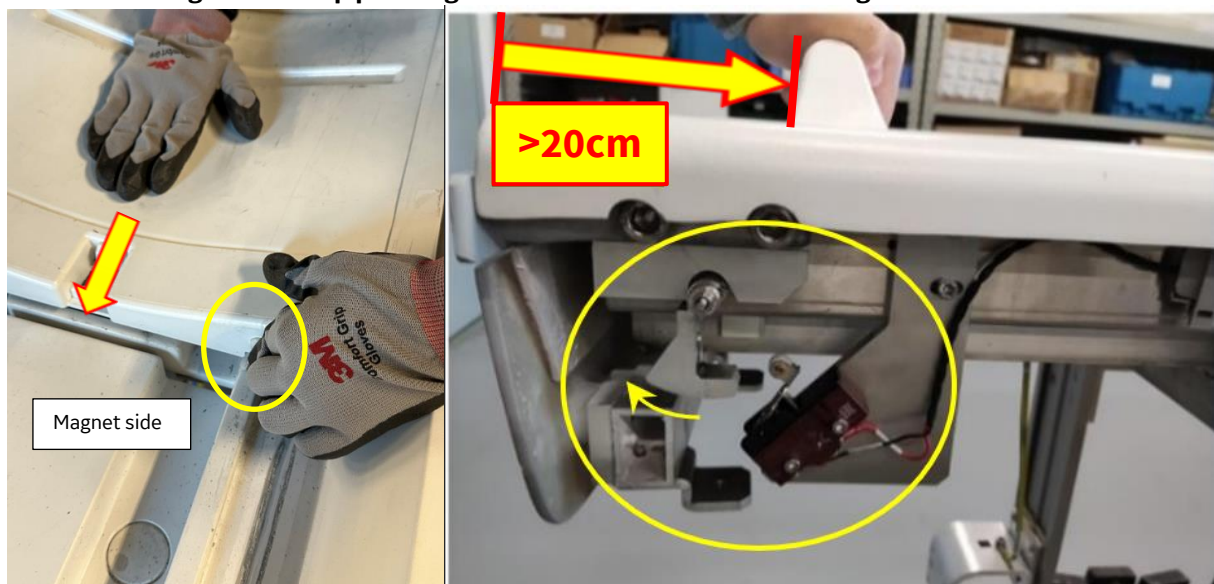


NOTE: If the cradle cannot be locked by the side latch, remove one shim, and repeat this step, if no more shims can be removed, adjust the side latch horizontally.

After the cradle can be locked, go to **Step.22.** and remove the sticker.

21. Keep pressing the button (on cradle magnet end) to release cradle from side latch, then push the table away toward the magnet side for more than 20cm, make sure the home sensor is released.

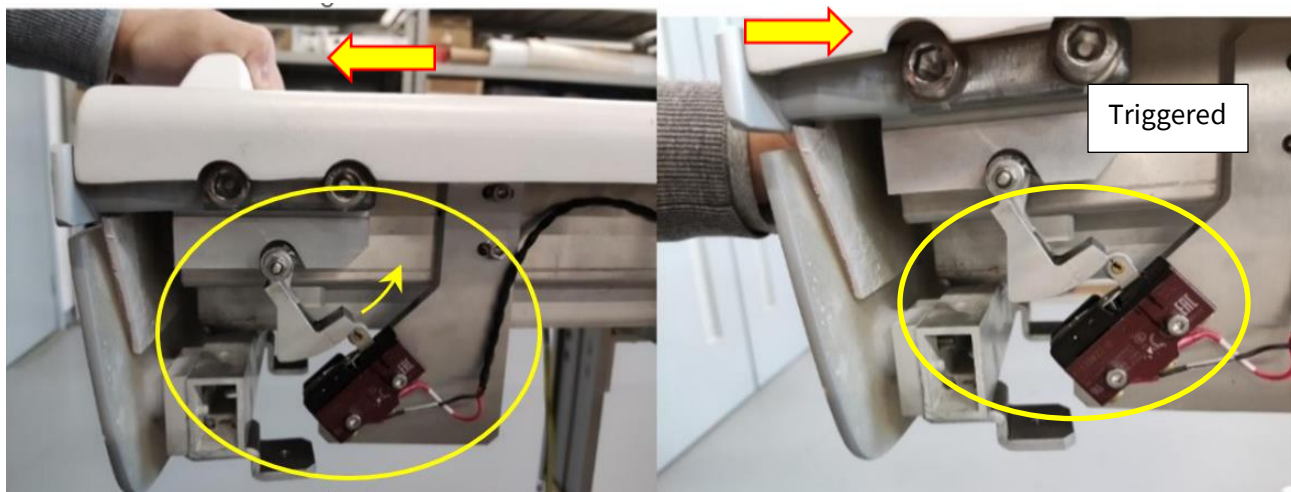
Figure 19. Keep pressing the Button to move cradle to magnet side > 20cm



Then pull the cradle back until it hit the bumper, make sure home sensor is triggered (a click sounds) then push the cradle to let it lock by side latch, make sure the home sensor is still triggered (no longer click sound).

NOTE: If the home sensor requests failed, adjust the sensor mount (see **Step.17**) until home sensor can be triggered/released (a click sounds) as requested.

Figure 20. Make sure the home sensor is in a suitable position



22. Make sure the cradle back to its rear end, home sensor is triggered and then push the cradle to check it locked by the side latch.

- a. Stick the same type of sticker (**8311142-1 Position Measurement Sticker of LHFT**) used in **Step.18** between cradle and Tabletop near the magnet side.

NOTE: Place the sticker parallel to the magnet end of the table.

- b. Then cut the middle of sticker and release the cradle from side latch (by keep pressing and holding the button on the magnet end of the cradle see **Figure 22**), then move the cradle forward until red arrows reached red line (which means move 4mm).
- c. Check during the movement whether the click sound happens (home sensor released), if heard the click sound, finish the process, if not, adjust the position of the sensor mount (**Step.17**) and go back to **Step.21**.

Figure 21. Use the sticker to calibrate the cradle position

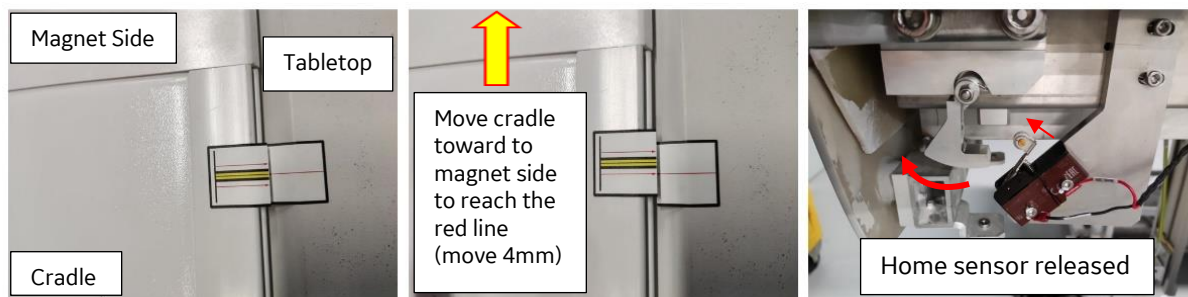


Figure 22. Press the button on the end of cradle & move the cradle to reach the middle red line



After the cradle can be locked, remove the sticker.

Close fixers and reinstall tabletop panels:

23. Put the home sensor cable back into the fixer and close all the fixer.

NOTE: If you find that the wires are not long enough for all the fixers to fix them, loosen the one or two closest fixers to the home sensor.

Figure 23. Fix the cable of home sensor



24. Fix back the Bottom panels by properly inserting the Side and Rear bumpers to Panel and FRP.

Figure 24. Fix back the Bottom panels



25. Tighten the Screws of the Tabletop panels.

Figure 25. Tighten the Screws



Please refer to below steps from Service Manual:

- 26. Removing LOTO – ISC.
- 27. Installing scissor covers.
- 28. Installing dock cover assembly.
- 29. Installing FRP bottom covers.
- 30. Checking patient table function.

Revision	Version	Date	Author	Description
01	01	19-Mar-2024	Qin Yu	Initial Release

End of Document